

Teaching Exam

Subject – Computer Science

**Topic – Computer Organization &
Architecture**

Lecture No.- 12



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Topics to be Covered

Topic



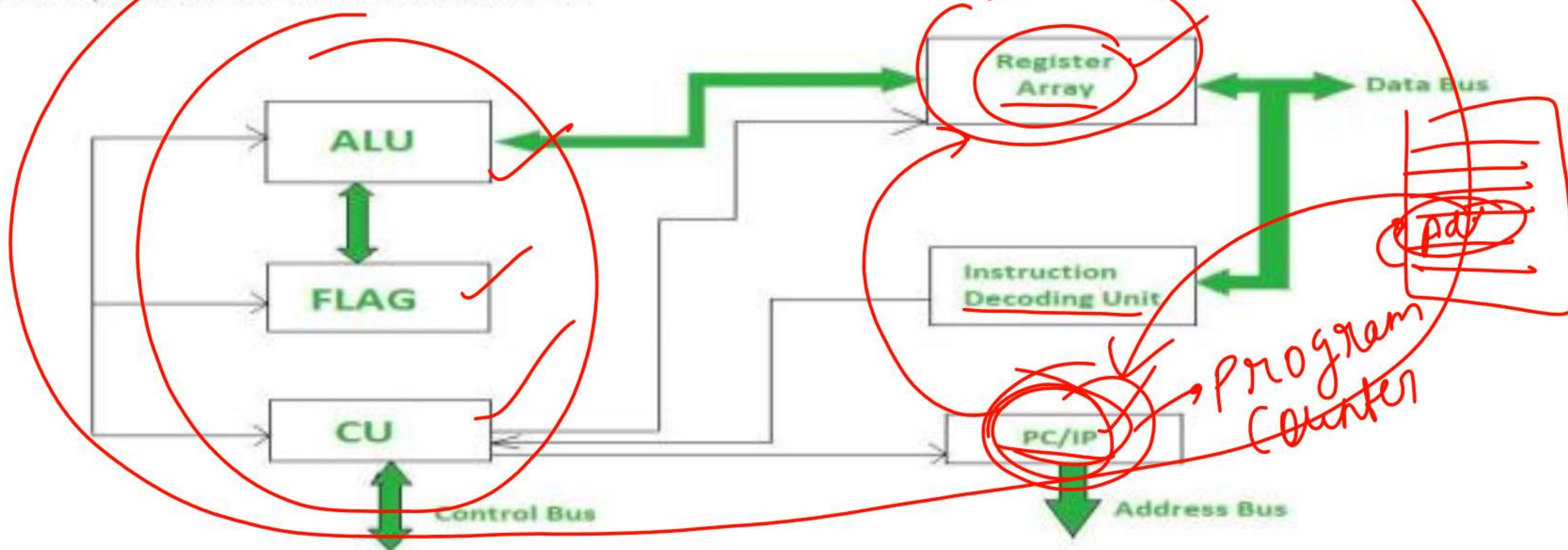
- 1:- Microprocessor – Uses, workflow, Application ✓
- 2:- Microcontroller – Uses, workflow, Application ✓
- 3:- Difference Between Microprocessor and Microcontroller. ✓

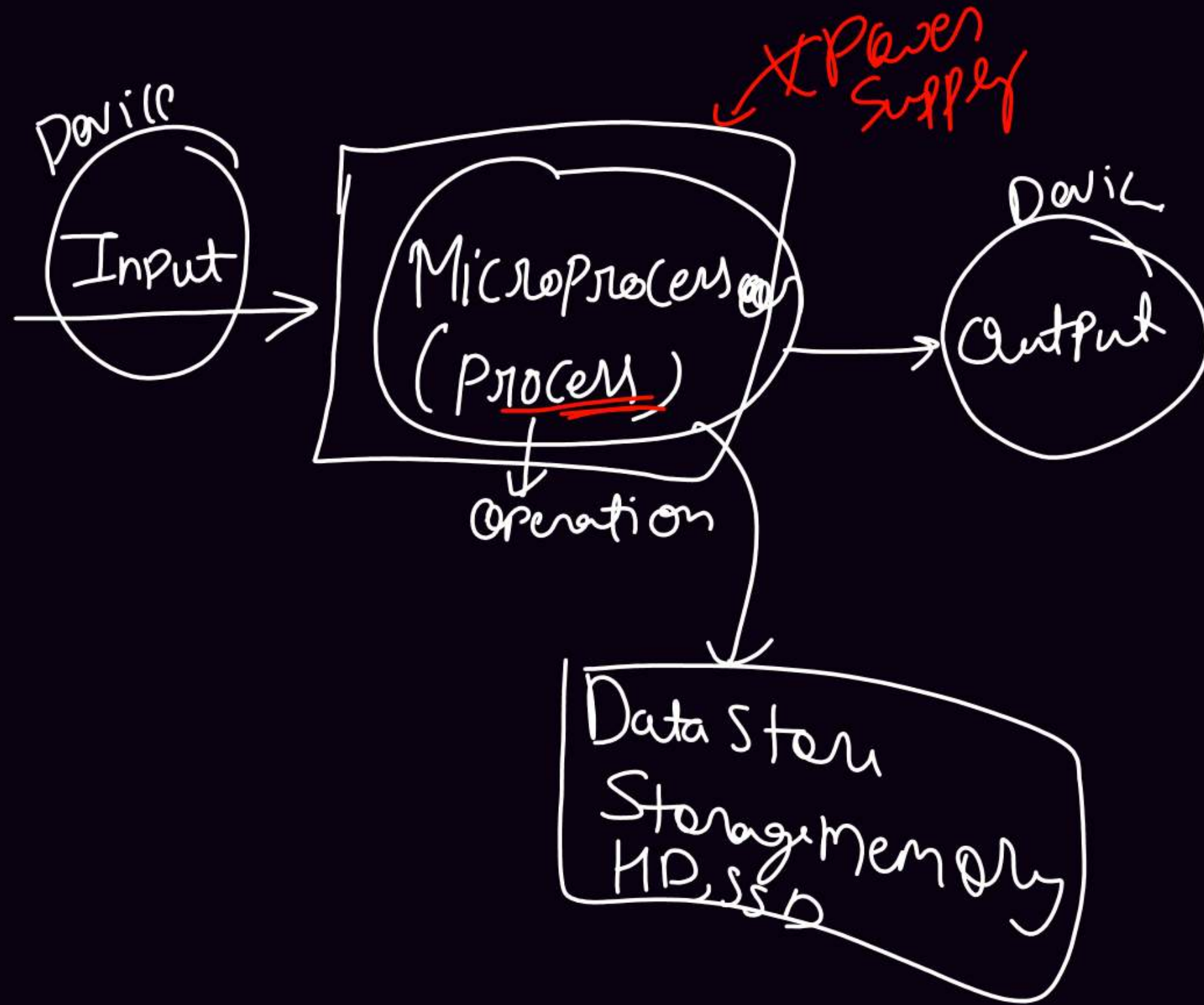


What is Microprocessor?

A Microprocessor is a programmable device that takes in input, performs some arithmetic and logical operations over it and produces the desired output. In simple words, a Microprocessor is a digital device on a chip that can fetch instructions from memory, decode and execute them, and give results. It is an important part of a computer architecture without which you will not be able to perform anything on your computer.

A typical Microprocessor structure looks like this.







Block Diagram of a Microprocessor

A Microprocessor takes a bunch of instructions in machine language and executes them, telling the processor what it has to do. The microprocessor performs three basic things while executing the instruction:

- It performs some basic operations like addition, subtraction, multiplication, division, and some logical operations using its Arithmetic and Logical Unit (ALU). New Microprocessors also perform operations on floating-point numbers.
- Data in microprocessors can move from one location to another.
- It has a Program Counter (PC) register that stores the address of the next instruction based on the value of the PC,
Microprocessor jumps from one location to another and makes decisions.



Basic Terms used in Microprocessor

Given below are the some basic term used in the microprocessors

- Instruction Set: Instruction set is a group of instructions in the microprocess which can be executed. It is the Interface between hardware and software.
- Bus: Buses are used to carry data, address and control information within a microprocessor. There are three types of Bus which are data buses, address buses, and control buses.
- IPC (Instructions Per Cycle): It is Measured as the number of instructions that a CPU can execute in a Single clock cycle.
- Clock Speed: It refers to the number of operations a processor can perform per second. It is Measured in megahertz (MHz) or gigahertz (GHz).
- Bandwidth: It is measured as is the number of bits processed during a single instruction.
- Word Length :It Shows the number of bits a processor can handle at a time.
- Data Types: It Supports various formats such as binary, ASCII, Signed and Unsigned Bits.

RA (R) (A) (L)
↓ ↓ ↓ ↓ ↓
8b 16b 16b 16b 8b 8b 8b 8b

1 byte = 8 bit



Evolution of Microprocessors

The Evolution of Microprocessors are categorized as

- **First Generation (4-bit Microprocessors):** The Intel Corporation first introduced the microprocessor in 1971-1972, naming it the Intel 4004. This initial microprocessor could perform basic arithmetic and logical functions. It also contained a control unit capable of fetching, decoding, and executing instructions.
- **Second Generation (8-bit Microprocessor):** The Intel Corporation first introduced the 8-bit microprocessor in 1973 with the Intel 8008. This was followed by an enhanced model, the Intel 8088.
- **Third Generation (16-bit Microprocessors):** The Third Generation introduced microprocessors capable of performing 16-bit operations. Introduced in 1978, this generation was represented by processors like the Intel 8086, Zilog Z800 and 80286.
- **Fourth Generation (32-bit Microprocessors):** The most used and famous 32-bit microprocessor was the Intel 80386.
- **Fifth Generation (64-bit Microprocessors):** Since 1995, microprocessors with 64-bit processors which feature 64-bit processors like Intel's Pentium Pro which allows multiple CPU's in one.



Types of Processor

Complex Instruction Set Computer (CISC)

CISC or Complex Instruction Set Computer is a computer architecture where instructions are such that a single instruction can execute multiple low-level operations like loading from memory, storing into memory, or an arithmetic operation, etc. It has multiple addressing modes within a single instruction. CISC makes use of very few registers.

Example of CISC are

- Intel 386 ✓
- Intel 486 ✓
- Pentium ✓
- Pentium Pro ✓
- Pentium II ✓
- Pentium III ✓
- Motorola 68000 ✓
- Motorola 68020 ✓
- Motorola 68040 etc. ✓



Reduced Instruction Set Computer (RISC)

RISC or Reduced Instruction Set Computer is a computer architecture where instruction is simple and designed to get executed quickly. Instructions get completed in one clock cycle this is because of the optimization of instructions and pipelining (a technique that allows for simultaneous execution of parts, or stages, of instructions more efficiently process instructions). RISC makes use of multiple registers to avoid large interactions with memory. It has few addressing nodes.

Example of RISC are

- IBM RS6000 ✓
- MC88100 ✓
- DEC Alpha 21064 ✓
- DEC Alpha 21164 ✓
- DEC Alpha 21264 ✓

CSIC > RSIC > EPIC
↓
i-generation

Explicitly Parallel Instruction Computing (EPIC) ✓

EPIC or Explicitly Parallel Instruction Computing permits computers to execute instructions parallel using compilers. It allows complex instructions execution without using higher clock frequencies. EPIC encodes its instruction into 128-bit bundles, each bundle contains three instructions which are encoded in 41 bits each and a 5-bit template field (contains information about types of instructions in a bundle and which instructions can be executed in parallel).

Example of EPIC is

- IA-64 (Intel Architecture-64)



Applications of Microprocessor

Given below are the Applications of the Microprocessors

- **Computer:** The Microprocessors are the CPU's in computers.
- **Embedded Systems:** It is utilized as the main processing block in the Embedded systems such as Washing machines, microwaves and other equipments.
- **Industrial Automation:** It can be used to control Industrial machinery and equipments.
- **Automotive:** The Modern day Vehicles uses Microprocessors in there ECUs.
- **Telecommunications:** The Microprocessors are utilized in the Telecommunications systems such as routers, modems, and switches.



Microcontroller

A microcontroller is a self-contained desktop that can be utilized in an embedded system. A few microcontrollers may run at clock rate rates and use four-bit expressions. Because many of the devices they control are battery-operated, microcontrollers must often be low-power. It is a small, low-cost computer built into a single chip. It is often used as part of an embedded system to control specific functions in electronic devices.

A microcontroller includes:

- A processor that runs instructions.
- Memory to store programs and data.
- Input/Output ports to connect with buttons, sensors, screens, etc.
- Communication ports (to connect with other devices).

Why Are Microcontrollers Useful?

- They are programmable, so you can make them do different tasks.
- Common programming languages used are C, C++, and assembly language.
- They are often low-power, which is great for devices that run on batteries.
- They are small, cheap, and perfect for controlling devices over a long time.
- Most microcontrollers are embedded, meaning they are built inside other devices.

Process
(limited
few specific
task)
+ Input Pin
+ Power Pin
+ Output Pin
+ ~~RAM~~ Storage memory

Microcontroller



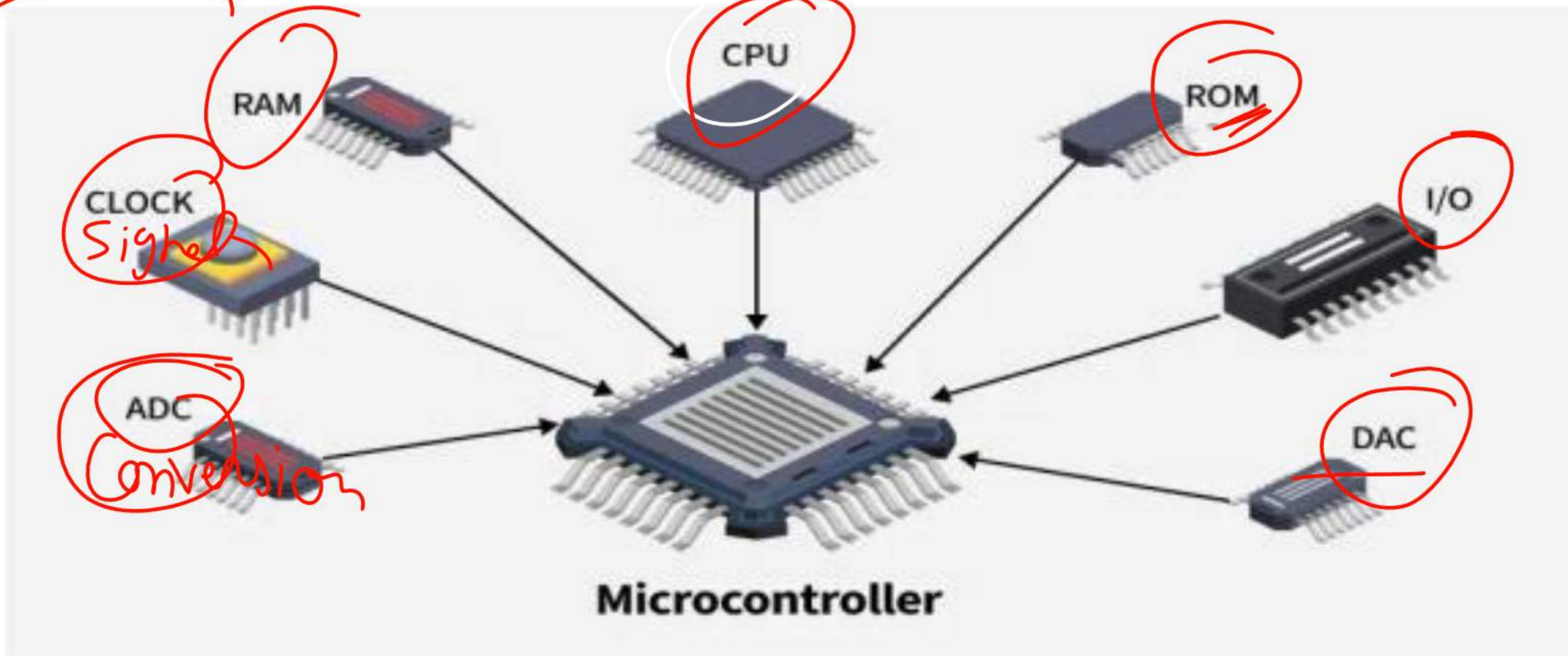
Topic: Microcontroller



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The microcontroller used in Embedded System. For example:

- Security Systems
- Laser Printers
- Automation System
- Robotics





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Working of Microcontroller

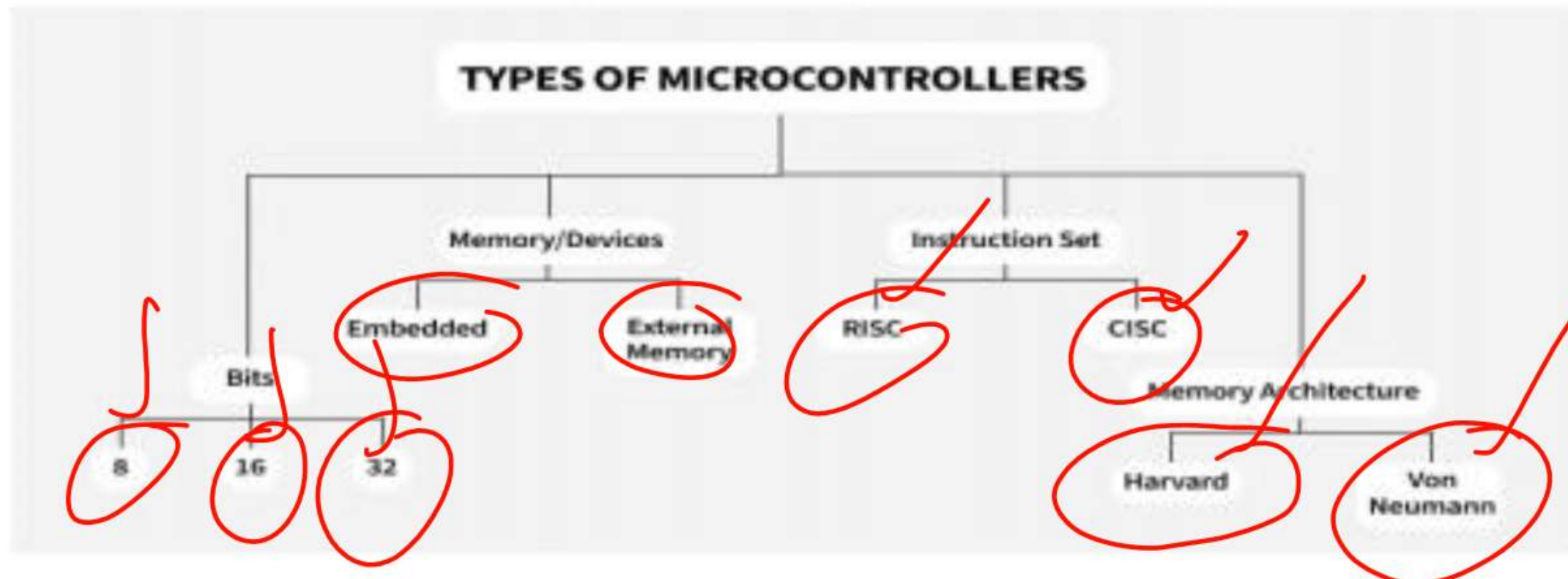
The microcontroller chip is a high-speed device, yet it is slow when compared to a computer. As a result, each command will be executed quickly within the microcontroller. The quartz oscillator is enabled and through control logic register once the supply is powered on. Parasite capacitors will be recharged for a few seconds while the early preparation is taking place.

Once the voltage level reaches its maximum value and the oscillator's frequency stabilizes, the operation of writing bits through special function registers becomes stable. Everything is controlled by the oscillator's CLK, and the whole electronics will begin to function. All of this happens in few nanoseconds.

A microcontroller's major role is that it can be thought of as a self-contained system with a processor memory. Its peripherals can be used in the same way that an 8051 microcontroller can. The bulk of microcontrollers in use today are embedded in other types of machinery such as telephones, appliances, vehicles, and computer system peripherals.

Types of Microcontroller

The types of microcontroller are characterized by the bits, memory architecture, memory/devices and instruction set.





Classification of Microcontrollers by Number of Bits

- **8-bit Microcontrollers**

- The internal bus is 8-bits wide.
- The ALU (Arithmetic Logic Unit) performs operations on 8 bits at a time.
- Example microcontrollers: Intel 8031/8051, PIC1x, Motorola MC68HC11.

- **16-bit Microcontrollers**

- The internal bus is 16-bits wide, providing better precision and performance than 8-bit.
- A 16-bit microcontroller can handle a wider range of numbers (0x0000 to 0xFFFF, or 0 to 65535) compared to the 8-bit range (0x00 to 0xFF, or 0 to 255).
- Example microcontrollers: Extended 8051XA, PIC2x, Intel 8096, Motorola MC68HC12.

- **32-bit Microcontrollers**

- These microcontrollers use 32-bit instructions for operations.
- Used in advanced applications like medical devices, engine control systems, office machines, and other embedded systems.
- Example microcontrollers: Intel/Atmel 251 family, PIC3x.



Classification of Microcontrollers by Memory Type

- Embedded Memory Microcontroller

- All functional blocks are built into the chip.
- Includes program memory, data memory, I/O ports, serial communication, counters, timers, and interrupts.
- Example: 8051 microcontroller (has everything on the chip).

- External Memory Microcontroller

- The microcontroller does not have all functional blocks on the chip.
- Some components, like program memory, need to be connected externally.
- Example: 8031 microcontroller (does not have program memory on the chip).

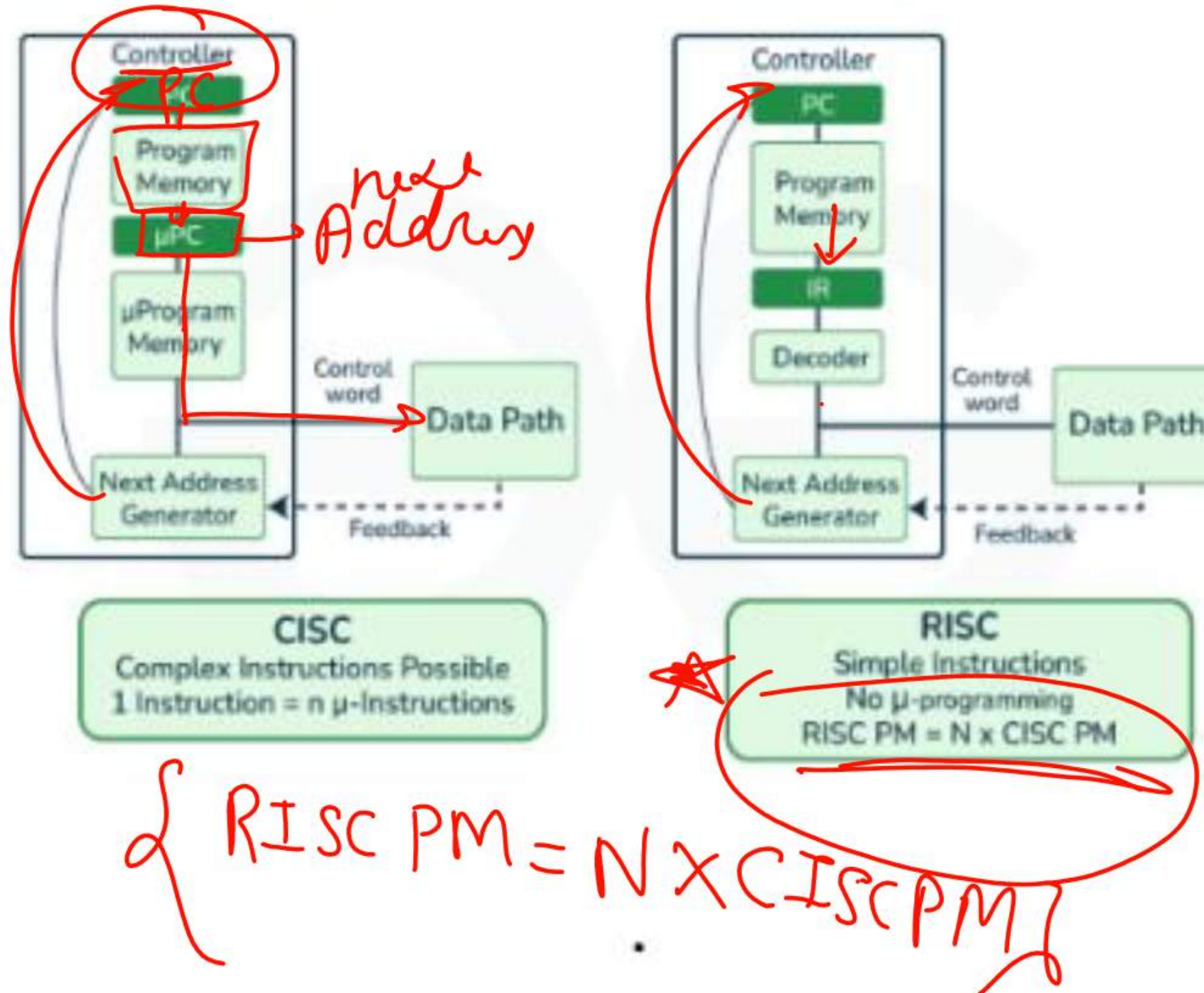
Classification of Microcontrollers by Instruction Set

- CISC (Complex Instruction Set Computer)

- Allows complex instructions that can do multiple tasks in one command.
- Reduces the need for many simpler instructions.
- Example: CISC systems shorten execution time by reducing the number of instructions in a program.

- RISC (Reduced Instruction Set Computer)

- Simplifies the instruction set, using fewer, simpler commands.
- Allows instructions to work on any register and access program and data simultaneously.
- Shortens execution time by reducing clock cycles per instruction.
- RISC systems generally offer better execution performance compared to CISC systems.





Classification of Microcontrollers by Memory Architecture

- Harvard Memory Architecture Microcontroller *Power*
 - Has separate memory for program and data.
 - Allows simultaneous access to both program and data memory, improving efficiency.
 - Common in microcontrollers that require faster data processing.
- Von Neumann Memory Architecture Microcontroller
 - Uses shared memory for both program and data.
 - Simplifies design but can be slower since the program and data share the same bus.

Key Characteristics of Von Neumann Architecture

1. **Single Memory for Data and Instructions:** Both data and program instructions are stored in the same memory.
2. **Shared Bus:** A single bus is used for transferring data, addresses, and control signals, which can limit performance.
3. **Sequential Execution:** Instructions are executed one at a time in a sequential manner.



Microcontroller Properties

- Microcontroller devices are capable of having words longer than 64 bits.
- Microcontroller consist of RAM , ROM , Timer , I/O Ports.
- Microcontroller ROM is used for program storage and RAM is used for data storage.
- It is designed by using CISC architecture.
- The power consumption of modern microcontrollers is significantly lower and have operating voltage range from 1.8V to 5.5V
- The latest feature of microcontroller is flash memory like EPROM and EEPROM.
- The most recent feature of a microcontroller is flash memory, such as EPROM and EEPROM.



Topic: Microcontroller

Uses of Microcontroller

Microcontrollers are used in a wide range of electronic devices and systems given below:

Everyday Uses of Microcontrollers:

- Devices that detect and control light (like automatic street lights).
- Devices that monitor and adjust temperature (like smart thermostats).
- Fire detection and safety systems (like smoke alarms).
- Tools used in factories to measure and monitor things.
- Systems that control industrial processes automatically.

Uses of Microcontrollers in Industrial Control:

- Devices that measure and track industrial data.
- Systems that manage and control production processes.

Uses of Microcontrollers in Measuring Devices:

- Voltage meters.
- Devices that count rotating parts (like in motors).
- Current meters.
- Portable measuring tools (like digital multimeters).



Issues in Microcontroller

Some of the most common issues that can arise with microcontrollers:

- **Timing Issues**
 - Microcontrollers need precise timing to run tasks correctly.
 - Any timing errors can cause malfunctions that are often hard to detect and fix.
- **Power Issues**
 - They need a stable power supply.
 - Power fluctuations or drops can cause the microcontroller to stop working or behave unpredictably.
- **Heat Issues**
 - Microcontrollers generate heat during use.
 - Too much heat can damage the chip or cause it to fail.
 - Caused by poor design, bad cooling, or high room temperatures.
- **Noise Issues**
 - Can be affected by electromagnetic interference (EMI) and radio frequency interference (RFI) from nearby devices.
 - This can lead to errors or unexpected behavior.
- **Code Issues**
 - The software or code running on the microcontroller may have bugs or errors.
 - This can lead to malfunctions or failure of the system.
- **Security Issues**
 - Microcontrollers can be hacked or infected with malware.
 - Risks include unauthorized access, data theft, or system control by attackers.



Topic: Difference between Microcontroller & Microprocessor



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Difference between Microprocessor and Microcontroller

The following table highlights all the important differences between microprocessors and microcontrollers ?

Parameter	Microprocessor	Microcontroller
Definition	Microprocessors can be understood as the heart of a computer system.	Microcontrollers can be understood as the heart of an embedded system.
What is it?	A microprocessor is a processor where the memory and I/O component are connected externally.	A microcontroller is a controlling device wherein the memory and I/O output component are present internally.
Circuit complexity	The circuit is complex due to external connection.	Microcontrollers are present on chip memory. The circuit is less complex.
Memory and I/O components	The memory and I/O components are to be connected externally.	The memory and I/O components are available.
Compact system compatibility	Microprocessors can't be used in compact system.	Microcontrollers can be used with a compact system.
Efficiency	Microprocessors are not efficient.	Microcontrollers are efficient.
Zero status flag	Microprocessors have a zero status flag.	Microcontroller doesn't have a zero status flag.
Number of registers	Microprocessors have less number of registers.	Microcontrollers have more number of registers.
Applications	Microprocessors are generally used in personal computers.	Microcontrollers are generally used in washing machines, and air conditioners.

THANK YOU